

# VAISHNAVI

Sr. Data Engineer

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## PROFESSIONAL SUMMARY:

- o Senior Data Engineer with 10+ years of experience delivering scalable data pipelines, ETL processes, and data architectures across cloud and on-premises environments.
- o Strong programming skills in Python, SQL, PowerShell, Bash, and Scala, with experience creating custom automation scripts and data transformation jobs.
- o Proven track record of creating robust cloud-based solutions, including working with GCP services such as Compute Engine, BigQuery, Dataflow, Pub/Sub, and IAM.
- o Hands-on experience developing and managing large-scale distributed systems on AWS (Lambda, EMR, Redshift, S3) and Azure (Data Lake, Blob Storage, Functions).
- o Expertise in relational and NoSQL databases, including MySQL, PostgreSQL, MongoDB, and SQL Server, with a deep understanding of data integration and performance optimization.
- o Adept at designing and maintaining high-performance data warehouses, ensuring optimal data ingestion, processing, and retrieval times.
- o Proficient in Git, Bitbucket, GitHub, JIRA, and Confluence for software development.
- o Expertise handling massive data sets using big data technologies such as Hadoop, HDFS, Apache Spark, Apache Kafka, and Apache Flink.
- o Extensive experience in implementing and managing data warehousing solutions with Snowflake, Redshift, and BigQuery for efficient data storage and querying.
- o Proficient in real-time data processing and streamlining workflows with Apache Beam, Spark Streaming, and Kafka to ensure high availability and low-latency data delivery.
- o Skilled in orchestrating complex data workflows using Apache Airflow and ADF to automate, monitor, and schedule ETL jobs for scalability and maintainability.
- o Advanced proficiency in machine learning models and predictive analytics, utilizing frameworks such as TensorFlow, PyTorch, and Scikit-learn to build and deploy intelligent systems.
- o Strong background in designing, implementing, and optimizing data pipelines and lakes within Google Cloud, AWS, and Azure environments.
- o Proficient in agile methodologies like Scrum and Kanban, ensuring smooth cross-functional collaboration and timely project delivery.
- o In-depth knowledge of data governance and data quality frameworks, ensuring data integrity and consistency across the enterprise.
- o Skilled in advanced data analytics and reporting using BI tools such as Looker, Tableau, Power BI, and custom SQL reports to provide actionable insights.
- o Expertise in managing data lifecycle, including extraction, transformation, storage, and consumption, focusing on scalability and high availability.
- o Strong problem-solving abilities, capable of analyzing complex issues, proposing innovative solutions, and driving continuous improvements in data architecture and performance.
- o Continuously improving performance by conducting code reviews, optimizing query performance, and managing system resources to meet performance goals.
- o Excellent communication and collaboration skills, proven to work effectively in cross-functional teams and foster strong stakeholder relationships.

## TECHNICAL SKILLS:

**Big Data Technologies:** Hadoop, HDFS, Hive, Apache Spark, Apache Beam, Apache Kafka, Apache Airflow, Apache Flink, Apache Sqoop, MapReduce, Spark SQL, PySpark, Spark Streaming

**Data Processing & ETL Tools:** Informatica, ADF, Talend, SSIS, AWS Glue

**Cloud Platforms:** Google Cloud Platform (GCP): Compute Engine, Storage, BigQuery, Dataflow, Pub/Sub, Logging, KMS, IAM, Kubernetes Engine (GKE); Microsoft Azure: Data Lake, Blob Storage, Functions, Active Directory, Synapse, HDInsight, DevOps; Amazon Web Services (AWS): EC2, S3, Lambda, RDS, EMR, Redshift, SQS, CloudWatch, IAM

**Databases & Data warehousing:** MySQL, PostgreSQL, SQL Server, MongoDB, DB2, Snowflake, Redshift, BigQuery

**Machine Learning & AI:** TensorFlow, PyTorch, Scikit-learn, Google Cloud AutoML, Predictive Analytics, Machine Learning Models, Advanced Analytics

**Programming Languages & Scripting:** Python, SQL, PowerShell, Bash, R, JavaScript, HTML5, CSS3, Scala, Shell Scripting

**Performance Optimization & Tuning:** Query Optimization, Data Pipeline Optimization, System Performance Tuning, Code Reviews, Database Performance Tuning, Resource Management, Latency

**Agile & Project Management:** Scrum, Kanban, Sprint Planning, Backlog Management, Cross-Functional Collaboration

**Data Modeling & Visualization:** Looker, Tableau, Power BI, Erwin Data Modeling

**Version Control & Collaboration:** Bitbucket, GitHub, JIRA, Confluence, SharePoint, Bugzilla

**Security & Compliance:** Google Cloud IAM, GCP KMS, Azure AD, Security Policies, Data Encryption, Identity and Access Management, Access Controls

**DevOps Tools & Practices:** Prometheus, Splunk, Google Stackdriver, Jenkins, Terraform, Azure DevOps

## EXPERIENCE:

**Credit Suisse, Morrisville NC | November 2023 - Present**

**Sr. Data Engineer**

### Roles & Responsibilities:

- o Designed ETL pipelines with Apache Beam and Google Cloud Dataflow for seamless data processing and transformation.
- o Architected scalable GCP infrastructure (Compute Engine, Cloud Storage, BigQuery) for efficient data access and storage.
- o Utilized Hadoop, Hive, and HDFS for large-scale data processing, ensuring high availability, scalability, and fault tolerance across data pipelines.
- o Utilized AutoML and Cloud AI Platform to improve machine learning capabilities.
- o Led data processing optimization efforts with PySpark and Spark SQL, significantly improving performance for complex data transformation tasks.
- o Optimized data ingestion using GCP Pub/Sub, Spark, and Dataflow.
- o Secured cloud resources and sensitive data using data security best practices with Google Cloud Key Management Service (KMS), ensuring compliance with industry standards.
- o Managed CI/CD pipelines and infrastructure automation with Jenkins, Terraform, and Docker, streamlining deployment workflows and improving efficiency.
- o Automated containerized applications with Google Kubernetes Engine (GKE), enabling dynamic scaling and improving resource management.
- o Monitored cloud infrastructure and application health using Stackdriver and Google Cloud Logging, ensuring system performance and quick issue resolution.
- o Optimized data pipelines by integrating Apache Pig for data transformations and improved business intelligence workflows.
- o Developed and maintained backup and disaster recovery strategies for critical data and systems using GCP tools, ensuring business continuity.
- o Implemented version control and collaborative workflows using Git, enabling smooth team collaboration and maintaining a history of changes.
- o Collaborated with teams using Agile and Scrum methodologies, ensuring consistent features and data solutions delivery according to business requirements.
- o Created interactive dashboards and reports using Google Data Studio, providing actionable insights to stakeholders for data-driven decision-making.
- o Optimized database performance by managing Cloud SQL and Bigtable, ensuring high availability, scalability, and security for structured and unstructured data.
- o Leveraged Apache Spark to process large datasets, enabling distributed data analysis and supporting batch and real-time data pipelines.
- o Implemented data governance policies to ensure compliance and high data quality across systems, leveraging Google Cloud KMS for secure key management.
- o Implemented data validation to maintain data accuracy and consistency.
- o Fostered a DevOps culture by integrating data engineering practices with CI/CD pipelines, automating testing and deployment workflows.

- o Monitored and troubleshoot data pipelines and infrastructure in real-time, using GCP Pub/Sub and Google Cloud Logging to address issues promptly and minimize downtime.

- o Ensured consistent data access and protection through collaborative governance and security.

**Environment:** Apache Beam, Google Cloud Dataflow, GCP Pub/Sub, Google Cloud Key Management Service (KMS), Google Cloud Platform (Compute Engine, Cloud Storage, BigQuery), Hadoop, Hive, HDFS, Google Cloud AI Platform, AutoML, PySpark, Spark SQL, Jenkins, Terraform, Docker, Google GKE, Stackdriver, Pig, Git, Google Data Studio, Cloud SQL, Bigtable, Apache Spark.

## **Kaiser Permanente, Oakland CA | May 2021 - November 2023**

### **Data Engineer**

#### **Roles & Responsibilities:**

- o Optimized machine learning workflows with NumPy and Pandas, enabling enhanced data manipulation and analysis for business decision-making processes.
- o Ensured data quality and integrity by applying data validation techniques and utilizing tools like Cloud Dataprep to cleanse and enrich data before processing.
- o Managed security and access controls with GCP IAM, safeguarding sensitive data and ensuring compliance with organizational security standards.
- o Utilized Hadoop, HDFS, and Hive for large-scale data solutions.
- o Created scalable data models with Erwin, ensuring efficient data organization, storage, and access for reporting and analysis purposes.
- o Defined and implemented data modeling best practices for high data quality.
- o Designed and managed data pipelines using Google Cloud technologies, including Compute Engine, Cloud Spanner, and Google Cloud Storage, ensuring seamless data integration and movement across multiple platforms.
- o Developed machine learning models using TensorFlow to enable predictive analytics and optimized data processing workflows for business intelligence applications.
- o Streamlined infrastructure automation and CI/CD pipelines using Docker, Kubernetes, Jenkins, and Terraform, improving the deployment process and reducing manual intervention.
- o Integrated and maintained NoSQL and relational data sources like Firestore, BigQuery, and Cloud Spanner, enabling fast data retrieval and scalability across projects.
- o Implemented secure and efficient data governance practices with GCP IAM, ensuring role-based access and compliance across distributed data systems.
- o Developed version-controlled data repositories using Git, ensuring smooth collaboration among team members while maintaining audit trails and history of changes.
- o Performed advanced data analytics using BigQuery, Firestore, and Google Analytics, delivering actionable insights for business teams and leadership.
- o Designed and deployed containerized applications using Docker and Kubernetes, enhancing scalability and reliability in a cloud-native environment.
- o Monitored system health and performance using Prometheus and Google Cloud Monitoring, enabling proactive issue resolution and optimized resource allocation.
- o Utilized Dataflow and Spark for high-velocity real-time data processing.
- o Implemented data transformation processes using Apache Pig, enriching data for business intelligence and advanced analytics projects.
- o Automated ETL processes leveraging Apache Spark and Google Dataflow, enabling real-time data processing for continuous data transformation and analytics.

**Environment:** NumPy, Pandas, Cloud Dataprep, GCP IAM, Hadoop, HDFS, Hive, Erwin, GCP, TensorFlow, Docker, Kubernetes, Jenkins, Terraform, Firestore, BigQuery, Google Analytics, Git, Prometheus, Google Cloud Monitoring, Google Dataflow, Apache Spark, Apache Pig.

## **Staples, Framingham MA | January 2019 - April 2021**

### **Data Engineer**

#### **Roles & Responsibilities:**

- o Designed and managed ETL workflows using Azure Data Factory (ADF), automating data integration processes across cloud and on-prem systems.
- o Built and optimized SQL queries for SQL Server, focusing on query optimization and ensuring the speed and scalability of data retrieval.
- o Applied machine learning algorithms with PyTorch for predictive analytics, providing actionable insights to support data-driven decision-making.

- o Designed and deployed cloud data warehouse solutions on Snowflake, integrating with Azure Databricks for big data analytics.
- o Utilized MongoDB and other NoSQL databases to handle unstructured data, ensuring flexibility and scalability in data storage.
- o Managed user access and data security using Azure Key Vault and Azure Active Directory, maintaining a secure data environment.
- o Leveraged Apache Spark and Hadoop for distributed data processing, ensuring efficient processing of large-scale datasets across cloud platforms.
- o Utilized Azure Blob Storage and Azure Data Lake to store and manage vast datasets and optimize data accessibility and security.
- o Implemented Apache Kafka for real-time data streaming, providing timely insights and event-driven processing.
- o Automated system processes with PowerShell and Azure Functions, improving efficiency and reducing manual tasks.
- o Proactively identified system issues with Azure Log Analytics.
- o Developed interactive reports and dashboards using Power BI, enabling stakeholders to visualize data trends and make informed decisions.
- o Automated deployments and ensure code quality with Azure DevOps CI/CD.
- o Conducted system monitoring and performance tuning to ensure optimal use of cloud infrastructure, reducing latency and improving reliability.
- o Led the implementation of event-driven architectures with Azure Functions, enabling responsive, scalable, and efficient cloud-based applications.
- o Built real-time data pipelines for analytics and reporting, leveraging Azure HDInsight and Azure Databricks for seamless big data processing.
- o Integrated version control with Bitbucket, collaborating across teams to maintain code consistency and facilitate efficient deployment.

**Environment:** ADF, Apache Spark, Hadoop, Azure Blob Storage, Azure Data Lake, Apache Kafka, PowerShell, SQL Server, PyTorch, Snowflake, Azure Databricks, MongoDB, Azure Key Vault, Azure AD, Azure Log Analytics, Azure Functions, Azure HDInsight, Bitbucket, Power BI, Azure DevOps.

## **Express Scripts, St. Louis MO | February 2017 - December 2018**

### **Data Engineer**

#### **Roles & Responsibilities:**

- o Utilized Apache Hive to manage large datasets, ensuring efficient query performance and optimal data storage solutions.
- o Managed AWS infrastructure, including EC2, S3, RDS, and Lambda, to efficiently deploy and scale data applications.
- o Developed data visualizations, reports, and interactive dashboards using Looker, ensuring stakeholders could access clear, actionable insights.
- o Integrated data sources and maintained schemas for consistency and integrity, enabling clean and reliable data models.
- o Implemented data security best practices by enforcing access controls data encryption, and ensuring compliance with relevant standards.
- o Built and managed data integrations with Informatica and automated ETL processes for seamless data transformation and movement.
- o Collaborated in an Agile and Kanban environment to design scalable data architectures, using Git for version control and seamless collaboration.
- o Applied data architecture best practices to ensure high availability, fault tolerance, and scalability of data systems.
- o Architected and maintained data systems leveraging Apache Spark and Hadoop, processing structured and unstructured data for real-time analytics.
- o Worked with ServiceNow to track and resolve data-related incidents, ensuring smooth operations of data pipelines and systems.
- o Developed and optimized data pipelines with Apache Kafka for real-time data streaming, ensuring minimal latency and high throughput.
- o Tuned database performance and optimized queries for faster data retrieval and processing.
- o Used Python (NumPy, Pandas) for data manipulation and analysis, supporting data-driven decisions.

- o Leveraged Sqoop for data import/export between relational databases and Hadoop ecosystems, improving data mobility.
- o Maintained data governance policies and documentation using SharePoint.

**Environment:** Apache Hive, AWS (EC2, S3, RDS, Lambda), Looker, Informatica, Git, Apache Spark, Hadoop, ServiceNow, Apache Kafka, Python, NumPy, Pandas, Sqoop, SharePoint.

## **Techsoft Solutions, Hyderabad India | May 2013 - October 2015**

### **Data Engineer**

#### **Roles & Responsibilities:**

- o Administered and optimized DB2 databases, ensuring efficient data storage and fast retrieval to support business-critical applications.
- o Utilized machine learning algorithms with Scikit-learn for predictive analytics, driving actionable insights from large datasets.
- o Designed and optimized data pipelines using Apache Spark and Hadoop for efficient batch and real-time data processing.
- o Leveraged Python, Pandas, and NumPy to manipulate and cleanse data, improving data quality and ensuring accurate reporting.
- o Employed Excel for advanced data analysis, including pivot tables and VLOOKUP, to support reporting and decision-making.
- o Performed data quality checks and developed data quality metrics to monitor and enhance the integrity of processed data.
- o Collaborated with cross-functional teams to maintain database performance and ensure smooth integration across different platforms.
- o Applied data enrichment techniques to enhance raw data, providing enriched, actionable insights for analytics.
- o Managed and transformed large datasets by leveraging HDFS, Hive, and HBase for seamless integration and data movement across systems.
- o Ensured high standards for database administration, optimizing queries and storage for enhanced performance and scalability.
- o Utilized Python for automation tasks, optimizing data workflows, and improving operational efficiency.
- o Implemented ETL workflows to streamline data integration and transformation processes, ensuring data accuracy and consistency.
- o Worked with Bugzilla to track and manage data processing and pipeline performance issues.

**Environment:** DB2, Scikit-learn, Apache Spark, Hadoop, Python, Pandas, NumPy, Excel, HDFS, Hive, HBase, Bugzilla.